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MILITARY SPECIFICATION

SHELL, HOWITZER, 105-MM., M60; WP, FS, HD, AND H FILLING ASSEMBLY

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the WP, FS, HD, and H filling assembly of the M60, 105-mm. howitzer shell without fuzing components.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawing, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU–P–236 — Paper, Filter, Analytical.

MILITARY

JAN–P–215 — Phosphorus, White.

JAN–P–253 — Paper, Liquid-Vesicant-Detector, M6.

JAN–C–314 — Crayon, Vesicant (H or L) Detector, C14.

JAN–C–379 — Chlorosulfonic Acid — Sulfur Trioxide Solution (FS).

JAN–H–502 — Heximine (For Use in Ammunition).

MIL–M–12051 — Mustard (HD) Vacuum Distillation Process.

MIL–M–12054 — Mustard (H) Levinstein Process.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

DRAWINGS

ORDNANCE CORPS

75–14–273 — Shell, Chemical, 105-mm., M60 — Loading Assembly, Marking Diagram and Details.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1315

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2.2 **Other publications.** The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

INTERSTATE COMMERCE COMMISSION

49 CFR 71–78 — Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition—Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71–78 (Rev. 1950.")

3. REQUIREMENTS

3.1 **Materials and components.** All materials, components, and subassemblies used in the filling assembly shall conform to the specifications and drawings forming a part of this specification.

3.2 **Filling.**

3.2.1 *WP.* When WP filling is specified (see 6.1), the shell, substantially clean and free from foreign matter, shall be filled with molten white phosphorus conforming to Specification JAN–P–215. Whenever process water is used in WP filling lines, the water shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0 (see 3.3.1). The depth of water remaining on the surface of the WP in the filled and closed shell shall not exceed 1/8 inch when tested as specified in 4.4.1. The filling shall be solidified with the shell in a nose-up position.

3.2.2 *FS.* When FS filling is specified (see 6.1), the shell, dry and substantially free from foreign matter, shall be filled with chlorosulfonic acid-sulfur trioxide solution conforming to Specification JAN–C–379.

3.2.3 *HD.* When HD filling is specified (see 6.1), the shell, dry and substantially free from foreign matter, shall be filled with distilled mustard conforming to Specification MIL–M–12051.

3.2.4 *H.* When H filling is specified (see 6.1), the shell, dry and substantially free from foreign matter, shall be filled with Levinstein mustard conforming to Specification MIL–M–12054. Prior to closing the filled shell 12.8 ± 1.3 grams of hexamine conforming to Specification JAN–H–502 shall be added. No hexamine shall be permitted to remain on the press-fit surface in the nose of the shell.

3.3 **Filling height.** The shell shall be tested as specified in 4.4.1.

3.3.1 *WP.* The surface of the WP in the filled and closed shell shall be 0 to 13/16 inch below the lowest point of the burster well adapter when the shell is inspected in a vertical nose-up position while the temperature of the WP filling is 130 ± 10°F.

3.3.2 *FS, HD, and H.* The distance below the bottom of the burster well adapter to the surface of the agent in the filled and closed shell shall be within the limits of the applicable table when the shell is inspected in a vertical nose-up position at the corresponding temperature range specified therein.

TABLE I. *FS—Filling level in closed shell*

Agent temperature (°F.)	Distance from bottom of burster well adapter to filling (inches)	
	Minimum	Maximum
40–49.9	3/4	1 1/2
50–59.9	1 1/16	1 7/16
60–69.9	5/8	1 7/16
70–79.9	9/16	1 3/8
80–89.9	1/2	1 5/16
90–99.9	7/16	1 5/16

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TABLE II. *HD and H—Filling level in closed shell*

Agent temperature (°F.)	Distance from bottom of burster well adapter to filling (inches)	
	Minimum	Maximum
60–69.9	5/8	1 3/8
70–79.9	9/16	1 3/8
80–89.9	9/16	1 5/16
90–99.9	1/2	1 5/16

3.4 **Assembly.** Closing, coating, marking, and assembly of the shell shall be in accordance with Drawing 75–14–273 except that the assembly shall not include the following: burster charge assembly, fuze well cup, and pad. Coating and marking for HD filled shell shall be the same as that for H filled shell except that the symbol HD shall be substituted for H.

3.5 **Cleaning after closing.** The exterior surface of the filled and closed shell and the interior of the burster casing shall be cleaned thoroughly to remove any grease, oil, agent, or other contamination.

3.6 **Leakage.** The cleaned shell shall not leak when tested as specified in the applicable paragraph of 4.4.2.

3.7 **Pilot lot.** A pilot lot shall be produced in accordance with 4.1.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.

4.1.1 *Pilot.* Prior to the start of regular production, and following any change in regular production methods, materials or equipment, a pilot lot of 225 shell shall be filled using the same methods, materials and equipment as will be employed during regular production.

4.1.2 *Regular production.* A lot shall consist of shell filled by the same filling installation under essentially the same conditions. However, not more than one lot of shell bodies shall be represented in any one lot of filled shell. The inspection lot for nondestructive inspection, within the above limitations, shall be designated by the inspector. A sublot shall consist of shell having the same weight zone designation (see 5.1). When specified in the contract or order, groups for shipment shall be formed from each production lot. The number of units in the group and the marking shall be as specified (see 6.1).

4.2 **Sampling for nondestructive inspection.** Sampling shall be conducted in accordance with the classification of defects (4.3.3) and Standard MIL–STD–105.

4.3 **Inspection.**

4.3.1 *Materials and components.* The inspector shall ascertain that all materials, components, and subassemblies of the shell, covered by separate drawings, specifications, or standards, are in accordance with the applicable drawings, specifications, or standards prior to final acceptance of the filled assembly.

4.3.2 *Packing and marking.* The inspector shall ascertain that the packing and marking of the shell conform to this specification.

4.3.3 *Classification of defects.* Inspection of the shell shall be conducted as follows:

4.3.3.1 *Shell, chemical, 105-mm., M60 — loading assembly.* (Dwg. 75–14–273)

Categories and defects	*Acceptance standards*
Critical	
A. Leakage	4.4.2
B. Agent contamination	
Major *AQL 1.0 percent defective*	
1. Burster casing not seated	
2. Threads damaged or contaminated	
3. Bourrelet diameter (maximum) after coating incorrect	
4. Filling excessive	4.4.1
5. Excess water in WP-filled shell	4.4.1
6. Marking (identification) incorrect	
7. Coating missing, inadequate, or incorrect	
8. Weight zone marking incorrect	

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Categories and defects	*Acceptance standards*
Minor *AQL 2.5 percent defective*	
51. Rotating band damaged	
52. Marking (identification) missing or illegible	
53. Filling insufficient	4.4.1
54. Threads not coated	
Incidental *AQL 10 percent defective*	
101. Contamination (other than filling agent)	

4.4 **Tests.**

4.4.1 *Filling level.* The shell shall be radiographed (see 6.1).

4.4.2 *Leakage.*

4.4.2.1 *WP.* Following the final cleaning operation and while the filling is still molten, the filled and closed shell shall be placed in a nose-down position or rolled on its side for a period of not less than 1 minute. The shell shall be observed for the emission of smoke which shall constitute leakage.

4.4.2.2 *FS.* Following the final cleaning operation, the shell shall be placed in a nose-down position or rolled on its side for a period of not less than 1 minute. The shell shall be observed for the emission of smoke which shall constitute leakage.

4.4.2.3 *HD or H.* After the final cleaning operation, a disk shall be inserted into the fuze well so that the coated side of the paper is in contact with the burster well adapter. The disk shall be cut either from detector paper conforming to Specification JAN–P–253 or from filter paper conforming to type I, class 2 of Specification UU–P–236. The filter paper shall be coated with crayon conforming to Specification JAN–C–314. The shell shall be maintained on its side or in a nose-down position at a temperature of 70°F. or higher for a minimum of 16 hours and the disk shall be examined for discoloration which shall be considered evidence of leakage.

4.5 **Pilot lot rejection criteria.** If any filled shell fails to meet the applicable drawing or specification requirements the pilot lot shall be rejected. The cause of failure shall be ascertained, corrective action instituted, and another pilot lot produced and subjected to the same examination and testing as for the initial pilot lot. No regular production lot shall be accepted prior to approval of the pilot lot.

5. PREPARATION FOR DELIVERY

5.1 **Packing, level A.** Unless otherwise specified, the filled and closed shell shall be packed for shipment or storage in the same manner as the empty shell were received provided the pack fulfills the Code of Federal Regulations 49 CFR 71–78. Shell shall be segregated by lot for shipping purposes.

5.2 **Marking.** In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Ordering data.** Procurement documents should specify:

(a) Title, number, and date of this specification.

(b) Filling agent required.

(c) That the contractor shall not be required to furnish radiographic equipment.

(d) Shipment group size and marking (when required).

6.2 **Precautions.** Because of the hazardous nature of filling munitions with chemical agents, filling installations are advised to take all necessary precautions to protect personnel from injury while filling shell in accordance with this specification.

6.3 **Filling aid.** The following information is furnished as an aid to procurement and not as a criterion for filling.

6.3.1 *WP.* Shell filled to a height of $5\frac{9}{32}$ to $5\frac{21}{32}$ inches below the nose of the shell or

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with 3.9 ± 0.3 pounds of WP have been found to comply with 3.3 provided the shell meet the dimensional requirements of the applicable drawings.

6.3.2 *FS, HD, and H.* Shell filled with agent within the limits of the applicable table for the corresponding temperature range have been found to comply with 3.3 provided the shell meet the dimensional requirements of the applicable drawings. The limits of each table are expressed in inches from the nose of the shell to the upper surface of the agent before the burster casing is inserted in the shell.

6.3.2.1 *FS.* The FS filling in the shell will weigh 4.2 ± 0.3 pounds.

TABLE III. *FS—Filling level in open shell*

Agent temperature (°F.)	Distance from nose of shell to filling (inches)	
	Minimum	Maximum
40–49.9	5 13/16	6 3/16
50–59.9	5 25/32	6 5/32
60–69.9	5 3/4	6 1/8
70–79.9	5 23/32	6 3/32
80–89.9	5 11/16	6 1/16
90–99.9	5 5/8	6

6.3.2.2 *HD and H.* The HD filling in the shell will weigh 2.7 ± 0.2 pounds. The H filling in the shell will weigh 2.8 ± 0.3 pounds.

TABLE IV. *HD and H—Filling level in open shell*

Agent temperature (°F.)	Distance from nose of shell to filling (inches)	
	Minimum	Maximum
60–69.9	5 3/4	6 1/8
70–79.9	5 23/32	6 3/32
80–89.9	5 11/16	6 1/16
90–99.9	5 21/32	6 1/32

6.4 Decontamination.

6.4.1 *WP.* Shell may be decontaminated by flushing with hot water while scrubbing vigorously with a wire brush.

6.4.2 *FS, HD, or H.* Shell may be decontaminated by washing with caustic solution and rinsing with hot water or steam. Other approved decontaminating agents may be used.

6.5 Safety classification. Shell filled in accordance with this specification carry the following Ordnance Safety Classification:

TABLE V. *Ordnance Safety Classification*

Agent	Chemical Ammunition Group	Explosive Hazard Class	Storage Compatibility Group
WP	C	2	A
FS	B	11	A
HD	A	11	A
H	A	11	A

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Custodians:
Army—Chemical Corps
Navy—Bureau of Ordnance
Air Force

Other interests:
Army—O
Navy—*MC*

Preparing Activity:
Army—Chemical Corps

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